Project Proposal

Egocentric Object and Action Memory with Calibrated Recall for People with Cognitive Impairment. CS 7641 Machine Learning, Fall 2026.

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HINDSIGHT: Egocentric Object and Action Memory with Calibrated Recall for People with Cognitive Impairment

CS 7641 Machine Learning, Fall 2026. Project Proposal.

Team: Jayit Saha, Veer Chheda, Aishwarya Daga, Shreya Dimri, Raghav Donakanti

Course section: Section B

Submission Date: 10/09/2026

1. Introduction and Background

Hindsight is a memory assistant for people with cognitive impairment. It watches first-person video, listens for questions, and keeps a timeline of objects and actions. Object retrieval finds the last frame where a requested object appeared. Put-down detection notices when an object leaves the hand and warns if the spot is unusual. Action detection labels what the user is doing and warns if it repeats within minutes. A spoken assistant answers timeline questions and says “I don’t know” when unsure.

Related work. Ego4D is a large first-person video dataset with labeled benchmarks for objects, language questions, actions, and hand-object state changes [1]. Xu et al. improved object finding by scoring candidates together [2]. SigLIP 2 maps images and text into one space, so a spoken description can be matched to crops [3]. Grounding DINO finds objects from a text prompt [4]; InternVideo2 gives the best video features for Ego4D [6]. Selective classification lets a model

refuse [9], scored by AURC [10].

Dataset. Ego4D episodic-memory and hands-and-objects subsets: object query crops and answer tracks (VQ2D), action segments in 110 classes (MQ), language questions with answer windows (NLQ), and "point of no return" (PNR) frames marking when an object leaves the hand, with official splits [1]. Ego4D ships precomputed features, so we decode only 60 hours of video.

2. Problem Definition

Problem. From first-person video recorded before a question, output the latest frame and box for a requested object; each put-down's time and how unusual its location is; action segments with labels and a repeat score; and a time window or refusal for a spoken question.

Motivation. Misplacing items and repeating the same question are among the most common signs of Alzheimer's dementia [12]. Catching the put-down or the repeat as it happens lets Hindsight warn before harm, not just answer afterward. Users who cannot crop an image must ask by voice, and because a user with memory loss believes whatever the assistant says, it must admit when unsure.

3. Methods

Data handling. Official Ego4D splits, one video per split. The refusal threshold is tuned on half of NLQ validation and reported on the other half; repetition pairs never cross splits.

Preprocessing. (1) Sample frames at 1 per second, 5 near PNR frames. (2) Find candidate crops with Grounding DINO, embed with SigLIP 2, reduce to 128-D with PCA [3], [4]. (3) Summarize the Shan et al. hand-object detector (touching or not, box speed) per 2-second window [5]. (4) Transcribe queries with Whisper large-v3 [11]. (5) Repetition pairs: same action class within 10 minutes is a repeat, over 30 minutes apart is not; time gap is excluded so the model cannot cheat.

Supervised learning. Retrieval: LogisticRegression, RandomForestClassifier, and MLPClassifier score whether a crop matches the query, against plain cosine similarity; spoken queries use SigLIP 2's text encoder, so no crop is needed. Put-downs: GradientBoostingClassifier predicts PNR windows and state change from hand-object features. Actions: MLPClassifier labels windows of precomputed InternVideo2 features [6], with ActionFormer as the reference [7]; a

pairwise MLP scores repetition against a same-class rule. Assistant: retrieval over detected objects and actions proposes a window; Qwen2.5-VL-7B checks its frames against the question and phrases the answer, refusing when the retrieval margin is low or the model disagrees [8], [9].

Unsupervised learning. KMeans indexes crop embeddings for faster retrieval. HDBSCAN groups crops into object identities and KMeans groups clip features into action types, checked against true labels with NMI. For misplacement, HDBSCAN clusters scenes around PNR frames and a smoothed count tracks where each object usually rests; a put-down in a rare place scores high.

4. Potential Results and Discussion

Metrics. Retrieval: Recall@1, MRR, VQ2D temporal and spatio-temporal AP. Put-downs: PNR timing error in seconds, state-change accuracy, misplacement AUROC. Actions: MQ mAP at overlaps 0.1 to 0.5; repetition AUROC and F1. Assistant: NLQ Recall@1 at 0.3 overlap, AURC, and coverage (questions answered) at 80% accuracy. Clustering: NMI, ARI.

Goals. A trained scorer beats cosine on Recall@1. PNR error matches or beats the Ego4D baseline. The action classifier lands within 5 mAP of ActionFormer on the same features. Repetition beats the same-class rule by 0.05 AUROC; misplacement beats simple counting. Answer checking raises coverage at 80% accuracy. The KMeans index costs under 2 Recall@1 points. Sustainability: precomputed features, 4-bit models.

Expected results. Trained scorers help with viewpoint changes and look-alike objects, not with missed detections. The assistant will answer under 30% of questions at 80% accuracy, since the best NLQ systems find the right window only 20 to 30% of the time. Demo footage will score lower.

5. References

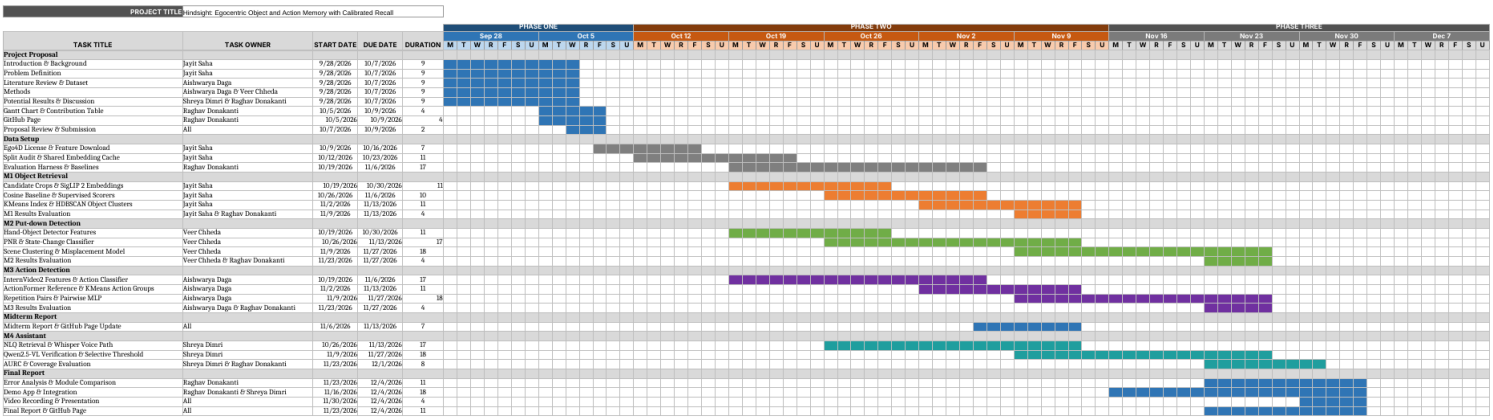
[1] K. Grauman et al., “Ego4D: Around the World in 3,000 Hours of Egocentric Video,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 18995-19012, doi: 10.1109/CVPR52688.2022.01842.
[2] M. Xu, Y. Li, C.-Y. Fu, B. Ghanem, T. Xiang, and J.-M. Pérez-Rúa, “Where Is My Wallet? Modeling

Object Proposal Sets for Egocentric Visual Query Localization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 2593-2603, doi: 10.1109/CVPR52729.2023.00255.
[3] M. Tschannen et al., "SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features," arXiv:2502.14786, 2025.
[4] S. Liu et al., "Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, LNCS vol. 15105, 2024, pp. 38-55, doi: 10.1007/978-3-031-72970-6_3.
[5] D. Shan, J. Geng, M. Shu, and D. F. Fouhey, "Understanding Human Hands in Contact at Internet Scale," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 9869-9878.
[6] Y. Wang et al., "InternVideo2: Scaling Foundation Models for Multimodal Video Understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, LNCS vol. 15143, 2024, pp. 396-416, doi: 10.1007/978-3-031-73013-9_23.
[7] C.-L. Zhang, J. Wu, and Y. Li, "ActionFormer: Localizing Moments of Actions with Transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, LNCS vol. 13664, 2022, pp. 492-510, doi: 10.1007/978-3-031-19772-7_29.
[8] S. Bai et al., "Qwen2.5-VL Technical Report," arXiv:2502.13923, 2025.
[9] Y. Geifman and R. El-Yaniv, "Selective Classification for Deep Neural Networks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 4878-4887.
[10] Y. Geifman, G. Uziel, and R. El-Yaniv, "Bias-Reduced Uncertainty Estimation for Deep Neural Classifiers," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
[11] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, vol. 202, 2023, pp. 28492-28518.
[12] Alzheimer's Association, "2025 Alzheimer's Disease Facts and Figures," *Alzheimer's Dement.*, vol. 21, no. 4, Art. no. e70235, 2025, doi: 10.1002/alz.70235.

Gantt Chart

Semester start (week 1) is Monday, September 28, 2026. Proposal due October 9, midterm report November 13, final report December 4. The three detection modules run in parallel from week 4 so the midterm report carries first results; the assistant and the demo follow once detections exist.

GANTT CHART



PROJECT TITLE	Hindsight: Egocentric Object and Action Memory with Calibrated Recall

TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION
Project Proposal				
Introduction & Background	Jayit Saha	9/28/2026	10/7/2026	9
Problem Definition	Jayit Saha	9/28/2026	10/7/2026	9
Literature Review & Dataset	Aishwarya Daga	9/28/2026	10/7/2026	9
Methods	Aishwarya Daga & Veer Chheda	9/28/2026	10/7/2026	9
Potential Results & Discussion	Shreya Dimri & Raghav Donakanti	9/28/2026	10/7/2026	9
Gantt Chart & Contribution Table	Raghav Donakanti	10/5/2026	10/9/2026	4
GitHub Page	Raghav Donakanti	10/5/2026	10/9/2026	4
Proposal Review & Submission	All	10/7/2026	10/9/2026	2
Data Setup				
Ego4D License & Feature Download	Jayit Saha	10/9/2026	10/16/2026	7
Split Audit & Shared Embedding Cache	Jayit Saha	10/12/2026	10/23/2026	11
Evaluation Harness & Baselines	Raghav Donakanti	10/19/2026	11/6/2026	17
M1 Object Retrieval				
Candidate Crops & SigLIP 2 Embeddings	Jayit Saha	10/19/2026	10/30/2026	11
Cosine Baseline & Supervised Scorers	Jayit Saha	10/26/2026	11/6/2026	10
KMeans Index & HDBSCAN Object Clusters	Jayit Saha	11/2/2026	11/13/2026	11
M1 Results Evaluation	Jayit Saha & Raghav Donakanti	11/9/2026	11/13/2026	4
M2 Put-down Detection				
Hand-Object Detector Features	Veer Chheda	10/19/2026	10/30/2026	11
PNR & State-Change Classifier	Veer Chheda	10/26/2026	11/13/2026	17
Scene Clustering & Misplacement Model	Veer Chheda	11/9/2026	11/27/2026	18
M2 Results Evaluation	Veer Chheda & Raghav Donakanti	11/23/2026	11/27/2026	4
M3 Action Detection				
InternVideo2 Features & Action Classifier	Aishwarya Daga	10/19/2026	11/6/2026	17
ActionFormer Reference & KMeans Action Groups	Aishwarya Daga	11/2/2026	11/13/2026	11
Repetition Pairs & Pairwise MLP	Aishwarya Daga	11/9/2026	11/27/2026	18
M3 Results Evaluation	Aishwarya Daga & Raghav Donakanti	11/23/2026	11/27/2026	4
Midterm Report				
Midterm Report & GitHub Page Update	All	11/6/2026	11/13/2026	7
M4 Assistant				
NLQ Retrieval & Whisper Voice Path	Shreya Dimri	10/26/2026	11/13/2026	17
Qwen2.5-VL Verification & Selective Threshold	Shreya Dimri	11/9/2026	11/27/2026	18
AURC & Coverage Evaluation	Shreya Dimri & Raghav Donakanti	11/23/2026	12/1/2026	8
Final Report				
Error Analysis & Module Comparison	Raghav Donakanti	11/23/2026	12/4/2026	11
Demo App & Integration	Raghav Donakanti & Shreya Dimri	11/16/2026	12/4/2026	18
Video Recording & Presentation	All	11/30/2026	12/4/2026	4
Final Report & GitHub Page	All	11/23/2026	12/4/2026	11

Semester start (W1) = Mon Sep 28, 2026. Proposal due Oct 9; midterm report Nov 13; final report Dec 4. Adjust due dates to the published course calendar.

Full chart with task dates and owners: Hindsight_GanttChart.xlsx

Contribution Table

Name	Proposal Contributions
Jayit Saha	Project idea and scope; Introduction and Problem Definition; object-retrieval methods; dataset research
Veer Chheda	Project idea and scope; Introduction and Problem Definition; object-retrieval methods; dataset research
Aishwarya Daga	Project idea and scope; Introduction and Problem Definition; object-retrieval methods; dataset research
Shreya Dimri	Project idea and scope; Introduction and Problem Definition; object-retrieval methods; dataset research
Raghav Donakanti	Project idea and scope; Introduction and Problem Definition; object-retrieval methods; dataset research

CS-7641-GROUP50-HINDSIGHT is maintained by vchheda7.

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